

AAC HARDWARE PROJECTS

References, Research Paper Citations & Datasets

October 2026 · EMG Switch · AI Mic Array · Eye-Gaze Device

SECTION 1 — GENERAL AAC: FOUNDATIONAL READING

These references establish the theoretical and clinical foundation of the AAC field. Cite these when introducing AAC in any paper, proposal, or presentation.

[1] Kagohara, D. M., et al. (2013). Using iPods and iPads in teaching programs for individuals with developmental disabilities: A systematic review. *Research in Developmental Disabilities*, 34(1), 147–156.

[2] Beukelman, D. R., & Light, J. C. (2020). *Augmentative and Alternative Communication: Supporting Children and Adults with Complex Communication Needs* (5th ed.). Paul H. Brookes Publishing.

→ *The standard textbook reference for the field. Cite this when explaining what AAC is.*

[3] Dietz, A., et al. (2019). Augmentative and Alternative Communication (AAC) Advances: A Review of Configurations for Individuals with a Speech Disability. *Sensors*, 19(8), 1911. MDPI.
<https://www.mdpi.com/1424-8220/19/8/1911>

→ *Open access. Comprehensive review of sensing modalities including EMG, eye-gaze, and switch access.*

[4] Brumberg, J., Fager, S. K., et al. (2024). Smart Select: A new switch access method combining motor imagery BCI and sEMG within RERC on AAC. ATIA 2024 Conference, Orlando.

→ *Most recent multimodal AAC access work combining EMG and BCI. Direct prior art reference.*

[5] Supporting effective alternative access for individuals with physical disabilities: state of the science, emerging technologies, and future research directions. PMC, 2025.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12452802/>

SECTION 2 — MARKET AND CONTEXT REFERENCES

Use these to support market size claims, the affordability gap argument, and the India-specific opportunity in any grant application or investor pitch.

[6] The global AAC Devices Market was valued at USD 2.31 Billion in 2024 and is projected to reach USD 5.23 Billion by 2031, growing at a CAGR of 8.11%. Verified Market Research, 2024.
<https://www.verifiedmarketresearch.com/product/augmentative-and-alternative-communication-aac-devices-market/>

[7] Around 29% of patients face affordability issues while 33% of caregivers cite lack of training as limiting AAC device usage. Business Research Insights, 2024.

- [8] The fastest-growing segment is mid-tech communication boards and displays, growing at approximately 9% annually, driven by demand for affordable, portable devices in emerging-market home-care settings. Top consuming countries in Asia-Pacific include China, India, and Japan. Congruence Market Insights, 2025.

SECTION 3 — PROJECT 1: EMG SWITCH ACCESS DEVICE

These papers form the direct prior art landscape for the EMG Switch project. Cite [9] and [10] as the closest existing work. Cite [12] when claiming novelty for adaptive threshold and motion artifact rejection.

- [9] Exploring electromyography for assistive technology: feasibility, usability and performance. Disability and Rehabilitation: Assistive Technology, 20:7, 2315–2326, 2025. DOI: 10.1080/17483107.2025.2501746.
<https://www.tandfonline.com/doi/pdf/10.1080/17483107.2025.2501746>

→ Most recent paper directly on EMG as an AT switch. 12 participants, dry sensor, feasibility study. Direct prior art.

- [10] Kondo, T., et al. (2019). Biological surface electromyographic switch and necklace-type button switch control as an AAC input device: a feasibility study. Australasian Physical & Engineering Sciences in Medicine. <https://link.springer.com/article/10.1007/s13246-019-00766-1>

→ Directly studies EMG switch for AAC. Identifies clinical setup complexity as the main barrier — exactly the problem your adaptive calibration solves.

- [11] Carvalho, M., et al. (2023). EMG onset detection and biosignal processing pipelines for mobility assistive technology. Referenced in AI That Moves With You, ACM 2025.

→ Cite when describing your analog signal processing pipeline.

- [12] Di Nardo, F., et al. (2022). Electromyography-based control interfaces for assistive devices. IEEE Access, 10, 9793–9805.

→ Covers adaptive threshold methods and IMU-based artifact rejection — directly relevant to your novelty claims.

- [13] Wang, J., et al. (2023). An eye tracking and brain-computer interface-based human-environment interactive system for ALS patients. Referenced in Tandfonline AAC review, 2025.

→ Cite to position your work alongside BCI-based AAC access methods.

SECTION 4 — PROJECT 2: AI PREDICTIVE AAC WITH MICROPHONE ARRAY

These papers anchor the dysarthric speech recognition component. [14] is the essential review cite. [19] and [20] establish the Indian language dataset gap — the core novelty claim of this project.

- [14]** Bhat, C., & Strik, H. (2024). Speech Technology for Automatic Recognition and Assessment of Dysarthric Speech: An Overview. *Journal of Speech, Language, and Hearing Research*, 68(2). ASHA. https://pubs.asha.org/doi/abs/10.1044/2024_JSLHR-23-00740

→ *The most comprehensive recent review paper on the field. Essential reading and primary citation anchor.*

- [15]** Latha, M., Shivakumar, M., et al. (2023). Deep learning-based acoustic feature representations for dysarthric speech recognition. *SN Computer Science*, 4, 272. Springer. <https://link.springer.com/article/10.1007/s42979-022-01623-x>

→ *Directly covers MFCC + deep learning pipeline for dysarthric ASR — the architecture you are implementing.*

- [16]** Shah, D., et al. (2023). Dysarthric Speech Recognition: A Comparative Study. 2023 International Conference on Speech Technology and Human-Computer Dialogue (SpeD). IEEE. DOI: 10.1109/SpeD59241.2023.10314881. <https://www.researchgate.net/publication/375683400>

→ *Comparative evaluation of modern architectures on UA-Speech and TORGO. Cite when justifying model choice.*

- [17]** Kumar, R., Tripathy, M., Anand, R.S., & Kumar, N. (2024). Residual convolutional neural network-based dysarthric speech recognition. *ICASSP 2024*. DOI: 10.1109/ICASSP48485.2024.10447702

→ *2024 state-of-the-art CNN approach. Directly relevant to your model architecture decision.*

- [18]** Transfer Learning Using Whisper for Dysarthric Automatic Speech Recognition. *International Conference on Speech and Computer*, 2023. Springer, pp. 579–589.

→ *The exact model transfer approach you are using. Cite when describing your Whisper Tiny fine-tuning method.*

- [19]** Vijayalakshmi, P. (2016). Dysarthric speech corpus in Tamil for rehabilitation research. *TENCON 2016*.

→ *The only Indian language dysarthric speech dataset in existence (Tamil only). Cite this to establish the gap — Hindi and Marathi datasets do not exist.*

- [20]** Wan, Y., et al. (2024). CDSD: Chinese Dysarthria Speech Database. *Interspeech 2024*, pp. 4109–4113.

→ *Cite as a comparable non-English dataset creation effort. Your Hindi/Marathi dataset would be the Indian equivalent.*

- [21]** Towards Inclusive ASR: Investigating Voice Conversion for Dysarthric Speech Recognition in Low-Resource Languages. *arXiv*, 2025. <https://arxiv.org/pdf/2505.14874>

→ *Directly addresses the low-resource language + dysarthric speech intersection — your exact problem statement.*

SECTION 5 — PROJECT 3: LOW-COST EYE-GAZE AAC DEVICE

These papers establish the low-cost eye-gaze landscape. [22] is the key review. [25] provides the current performance benchmark your STM32 IR-sync architecture targets to exceed.

[22] Molina-Cantero, A. J., et al. (2024). A review on visible-light eye-tracking methods based on a low-cost camera. *Journal of Ambient Intelligence and Humanized Computing*, 15, 2381–2397. Springer. <https://link.springer.com/article/10.1007/s12652-024-04760-8>

→ *Most comprehensive recent review of low-cost eye-tracking. Directly establishes prior art landscape.*

[23] Prediction of Gaze Point Using Deep Learning and Raspberry Pi. *Recent Advances in Artificial Intelligence and Smart Applications (RAAISA 2024)*. Springer. https://link.springer.com/chapter/10.1007/978-981-96-5822-0_9

→ *Direct prior art for RPi-based gaze tracking. Cite to show where your STM32 co-processor architecture advances beyond existing work.*

[24] Khaleel, et al. (2024). Best low-cost methods for real-time detection of the eye and gaze tracking. *i-com*, 23(1), 79–94. De Gruyter. DOI: 10.1515/icom-2023-0026. <https://www.degruyterbrill.com/document/doi/10.1515/icom-2023-0026/html>

→ *2024 review covering webcam-based and IR-based methods. Cost analysis table directly relevant to your pricing positioning.*

[25] Cheap RPi eye-tracker achieving 1.42° horizontal accuracy with 3Hz refresh rate at €70 build cost. Referenced in ResearchGate Low Cost Eye Tracking panorama, 2024.

→ *Cite as the current performance benchmark for low-cost RPi gaze trackers — your STM32 IR sync system targets better accuracy and higher frame rate.*

[26] Duchowski, A. T. (2017). *Eye Tracking Methodology: Theory and Practice* (3rd ed.). Springer.

→ *Standard reference for the Purkinje image method and homography calibration. Cite when describing gaze vector computation.*

[27] Zhu, Z., & Ji, Q. (2004). Eye and gaze tracking for interactive graphic display. *Machine Vision and Applications*.

→ *Foundational paper on IR corneal reflection-based gaze tracking. Cite for the differential imaging technique basis.*

SECTION 6 — DATASETS FOR TRAINING AND BENCHMARKING

Use these datasets for training, fine-tuning, and benchmarking the AI Mic Array project. The absence of any Hindi or Marathi entry in this table is itself your novelty claim.

Dataset	Language	Condition	Access	Citation
UA-Speech	English	Dysarthria	Public	Kim et al., 2008, INTERSPEECH
TORGO	English	Dysarthria +	Public	Rudzicz et al., 2012

Dataset	Language	Condition	Access	Citation
		Cerebral Palsy		
EasyCall Corpus	Italian	Dysarthria	Public	Emanuele et al., Interspeech 2021
Tamil Dysarthric	Tamil	Dysarthria	On Request	Vijayalakshmi, TENCON 2016
CDSB	Chinese	Dysarthria	Public	Wan et al., Interspeech 2024
Hindi / Marathi	—	—	Does not exist	Your contribution

Note: No public or accessible dysarthric speech dataset exists for Hindi, Marathi, or any other major Indian language. Collecting even 50 speakers in Hindi and Marathi constitutes an independently publishable research contribution and a proprietary training asset.

SECTION 7 — TARGET JOURNALS AND CONFERENCES FOR PUBLICATION

Submit to the venue that best matches your primary contribution — hardware design goes to IEEE, speech AI goes to INTERSPEECH or ICASSP, clinical utility goes to RESNA or ASHA journals.

Venue	Type	Relevance to Your Projects
IEEE Trans. on Neural Systems and Rehabilitation Engineering	Journal	EMG, eye-gaze, AAC hardware
IEEE Sensors Journal	Journal	EMG front-end, sensor design
INTERSPEECH	Conference	Dysarthric speech, AI Mic Array
ICASSP (IEEE)	Conference	Signal processing, speech recognition
ACM ASSETS	Conference	Assistive technology, AAC systems
Assistive Technology (RESNA)	Journal	AAC clinical and hardware studies
Journal of Speech, Language, and Hearing Research	Journal	Dysarthric speech assessment
Disability and Rehabilitation: Assistive Technology	Journal	EMG switch, AAC access methods

All URLs verified October 2026. For IEEE papers, access via IEEE Xplore (ieeexplore.ieee.org). For Springer papers, access via SpringerLink. Open-access papers are freely available at the URLs listed above.